

Frameworks Used

- Hugging Face Transformers: Used as the high-level library for loading, configuring, and running all pre-trained models in the system.
- PyTorch: Acts as the core deep learning framework responsible for tensor operations, model execution, and performance optimization.

Models Used and Justification

Embedding Model: CLIP (openai/clip-vit-base-patch32)

- Used in `embedding_engine.py` to generate vector embeddings for both text chunks and images.
- Maps text and images into a shared vector space, enabling direct similarity comparison using cosine similarity.
- Supports multimodal retrieval by allowing text queries to match relevant images without additional alignment models.
- Provides strong zero-shot learning capabilities, eliminating the need for task-specific fine-tuning.
- Widely adopted industry standard with strong community support and a balanced trade-off between accuracy and efficiency.

Alternatives Considered for Embeddings

- ResNet / VGG (Image) + BERT (Text): Rejected due to incompatible embedding spaces requiring complex alignment training.
- SigLIP / ALIGN: Considered as modern alternatives, but CLIP was preferred for its maturity, tooling, and proven reliability.

Captioning Model: BLIP (Salesforce/blip-image-captioning-base)

- Used in `caption_engine.py` to generate natural language descriptions for images.
- Produces rich, coherent captions that capture objects along with their relationships.
- Enables text-based indexing of images, acting as a fallback or complement to vector-based retrieval.
- Delivers strong vision–language performance while maintaining an efficient base model size.

Alternatives Considered for Captioning

- GIT (Microsoft): A strong image-to-text model, but BLIP was preferred due to its focused captioning optimization.
- Florence-2: A powerful unified vision-language model, considered excessive for basic captioning requirements.
- YOLO (Object Detection): Rejected because it only identifies objects without capturing contextual or relational information.